

Session 13 - Class Diagram Sample

For the class diagram, I chose to combine my experience in sewing and wardrobe department management with programming. Some of the core basics in a sewing business include keeping track of customers, employees, and supply inventory. The class diagram I created provides a visual representation of how all of these components work together.

The solution I am providing uses seven classes to represent the key parts of the business. Starting off I created the Person class at the top, which is meant to carry information that is needed between both customers and employees like name, phone number, and email address. In this case, both the Customer and Employee classes will inherit the common needed basic information from the Person class. After this, they would include their own respective additional details on top of the inherited information. Under the Employee class, I have included a Seamstress class and a Manager class which would all inherit information from the Employee class and then include additional respective needed information for the two separate job titles. I included an Order class, which details information about each sewing job including things like fabric type, price, and due date. Finally, I created the Fabric class to manage the inventory of the supplies in the department.

The diagram contains two examples of regular inheritance, and one example of multiple inheritance. The two regular inheritance examples are the Customer class inheriting the information from the Person class, and the Employee class inheriting the information from the Person class. This is shown using solid lines with hollow triangles pointing up to the Person class. The Seamstress class is the example that uses multiple inheritance because it inherits information from both the Employee class and the Order class at the same time. This is shown by using a solid line and a filled in triangle triangle that point toward the Order box to show that the Seamstress is both an Employee and directly tied to the Orders she is working on.

See Diagram on next page!

Assignment Prompt:

Develop a class diagram for a problem of your choice. You may use Word to create the diagram or other software you find useful. You can download a free trial of Microsoft Visio or sign up for the software noted in the video. Lastly, you may draw out the diagram on paper and scan or take a picture of the diagram to submit for grading.

Your problem and diagram should include:

- 2-3 paragraphs on the problem you are addressing and scope of your solution.
- at least 6 objects.
- two examples of inheritance.
- one example of multiple inheritance.

Regular Inheritance: → Sewing Business Class Diagram
Multiple Inheritance: →

